MH1101 Tutorial 9 (Week 10)

Solution

Reference: Topic 5.1, 5.2 (Lecture Notes)

1. Use the Integral Test to determine whether the series is convergent or divergent. You need to explain why the corresponding function is decreasing on the domain under consideration.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(3n-1)^4}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^3}{n^4+4}$

(c) $\displaystyle\sum_{n=1}^{\infty} ne^{-n^2}$

Solution

(a) The function $f(x) = \frac{1}{(3x-1)^4}$ is positive, continuous on the interval $[1, \infty)$.

The function $f(x)$ is also decreasing on $[1, \infty)$ since

$$f'(x) = \frac{-12}{(3x-1)^5} < 0,$$

for all $x \in [1, \infty)$.

So the Integral Test applies.

Substituting $u = 3x - 1$ ($\Longrightarrow du = 3\,dx$) we have

$$\int \frac{1}{(3x-1)^4}\,dx = \int \frac{1}{u^4}\,\frac{1}{3}\,du = \frac{1}{3}\int u^{-4}\,du = \frac{u^{-3}}{-9} + C.$$

When $x = 1$, $u = 3(1) - 1 = 2$; and when $x = t$, $u = 3t - 1$. Thus,

$$\begin{aligned}
\int_1^\infty \frac{1}{(3x-1)^4}\,dx &= \lim_{t\to\infty}\int_1^t \frac{1}{(3x-1)^4}\,dx \\
&= \lim_{t\to\infty}\frac{1}{3}\int_2^{3t-1} u^{-4}\,du \\
&= \lim_{t\to\infty}\left[\frac{u^{-3}}{-9}\right]_2^{3t-1} \\
&= \lim_{t\to\infty}\left(\frac{(3t-1)^{-3}}{-9}-\frac{(2)^{-3}}{-9}\right) \\
&= \frac{1}{72}.
\end{aligned}$$

Since this improper integral is convergent, the series $\sum_{n=1}^{\infty}\frac{1}{(3n-1)^4}$ is also convergent by the Integral Test.

(b) The function $f(x)=\frac{x^3}{x^4+4}$ is clearly positive and continuous on $[1,\infty)$. The function $f(x)$ is decreasing on $[2,\infty)$ since

$$f'(x)=\frac{(x^4+4)(3x^2)-x^3(4x^3)}{(x^4+4)^2}=\frac{12x^2-x^6}{(x^4+4)^2}=\frac{x^2(12-x^4)}{(x^4+4)^2}<0$$

for all $x\geq 2$.

So we can apply the Integral Test on $[2,\infty)$.

$$\begin{aligned}
\int_2^\infty \frac{x^3}{x^4+4}\,dx &= \lim_{t\to\infty}\int_2^t \frac{x^3}{x^4+4}\,dx \\
&= \lim_{t\to\infty}\left[\frac{1}{4}\ln(x^4+4)\right]_2^t \\
&= \lim_{t\to\infty}\left(\frac{1}{4}\ln(t^4+4)-\frac{1}{4}\ln(20)\right) \\
&= \infty.
\end{aligned}$$

So by the Integral Test the series $\sum_{n=2}^{\infty}\frac{n^3}{n^4+4}$ diverges. This implies that the original series $\sum_{n=1}^{\infty}\frac{n^3}{n^4+4}$ also diverges.

(c) The function $f(x) = xe^{-x^2}$ is positive and continuous on $[1, \infty)$.
Notice that
$$f'(x) = e^{-x^2} - 2x^2e^{-x^2} = e^{-x^2}(1 - 2x^2).$$
So $f'(x) < 0 \iff 1 - 2x^2 < 0$. Thus, $f'(x) < 0$ if $x > \sqrt{\frac{1}{2}} \approx 0.7$. Thus, $f'(x) < 0$ if $x \geq 1$. That is, $f(x)$ is decreasing on $[1, \infty)$.
We can use the Integral Test on $[1, \infty)$.

$$\begin{aligned}
\int_1^\infty xe^{-x^2}\,dx &= \lim_{t\to\infty} \int_1^t xe^{-x^2}\,dx \\
&= \lim_{t\to\infty} \left[-\frac{1}{2}e^{-x^2}\right]_1^t \\
&= \lim_{t\to\infty} \left(-\frac{1}{2}e^{-t^2} + \frac{1}{2}e^{-1}\right) \\
&= \frac{1}{2}e^{-1}.
\end{aligned}$$

By the Integral Test, the series $\sum_{n=1}^\infty ne^{-n^2}$ converges.

□

2. Determine whether the series converges or diverges.

(a) $\displaystyle\sum_{n=1}^\infty \frac{1}{n^3+8}$

(b) $\displaystyle\sum_{n=1}^\infty \frac{n-2}{n\sqrt{n}}$

(c) $\displaystyle\sum_{n=1}^\infty \frac{9^n}{3+10^n}$

(d) $\displaystyle\sum_{n=1}^\infty \frac{n^2+n+1}{n^4+n^2}$

(e) $\displaystyle\sum_{n=1}^\infty \sin\left(\frac{1}{n}\right)$

(f) $\displaystyle\sum_{n=1}^\infty \frac{1}{n^{1+1/n}}$

(g) $\displaystyle\sum_{n=1}^\infty \frac{e^n+1}{ne^n+1}$

(h) $\displaystyle\sum_{n=1}^\infty \frac{\sqrt{1+n}}{2+n}$

(i) $\displaystyle\sum_{n=1}^\infty \frac{n+4^n}{n+6^n}$

(j) $\displaystyle\sum_{n=1}^\infty \frac{\tan^{-1} n}{n^{1.2}}$

Solution

(a) $\dfrac{1}{n^3+8} < \dfrac{1}{n^3}$ for all $n \geq 1$, so $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3+8}$ converges by comparison with $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3}$ which converges because it is a p-series with $p = 3 > 1$.

(b) Let $a_n = \dfrac{n-2}{n\sqrt{n}}$. Note that a_n is positive for $n \geq 3$.

Let $b_n = \frac{1}{\sqrt{n}}$. Then b_n is positive for $n \geq 1$.

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{\frac{n-2}{n\sqrt{n}}}{\frac{1}{\sqrt{n}}} \\
&= \lim_{n\to\infty} \left(1 - \frac{2}{n}\right) \\
&= 1.
\end{aligned}$$

By the Limit Comparison Test, $\sum_{n=3}^{\infty} a_n$ diverges since $\sum_{n=3}^{\infty} b_n$ diverges (it is a p-series with $p = 1/2 \leq 1$).

It follows that the series $\sum_{n=1}^{\infty} a_n$ also diverges.

(c) $\dfrac{9^n}{3+10^n} \leq \dfrac{9^n}{10^n}$ for all $n \geq 1$. So $\displaystyle\sum_{n=1}^{\infty} \frac{9^n}{3+10^n}$ converges by comparison with $\displaystyle\sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$ which is a convergent geometric series.

(d) Let $a_n = \dfrac{n^2+n+1}{n^4+n^2}$, and $b_n = \dfrac{1}{n^2}$. Both a_n and b_n are positive for all $n \geq 1$. Then

$$
\begin{aligned}
\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{n^2+n+1}{n^4+n^2}\cdot\frac{n^2}{1}\\
&= \lim_{n\to\infty}\frac{n^2+n+1}{n^2+1}\\
&= \lim_{n\to\infty}\frac{1+\frac{1}{n}+\frac{1}{n^2}}{1+\frac{1}{n^2}}\\
&= 1.
\end{aligned}
$$

Since $\sum_{n=1}^{\infty} b_n$ is convergent (it is a p-series with $p = 2 > 1$), by the Limit Comparison Test, the series $\sum_{n=1}^{\infty} a_n$ is also convergent.

(e) Let $a_n = \sin\left(\frac{1}{n}\right)$, and $b_n = \frac{1}{n}$. Both a_n and b_n are positive for all $n \geq 1$.

$$
\begin{aligned}
\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}\\
&= \lim_{x\to 0}\frac{\sin x}{x}\\
&= \lim_{x\to 0}\frac{\cos x}{1} \quad \text{(by L'Hospital Rule)}\\
&= \cos 0 = 1.
\end{aligned}
$$

The series $\sum_{n=1}^{\infty} b_n$ diverges since it is the Harmonic series. By the Limit Comparison Test, the series $\sum_{n=1}^{\infty} a_n$ also diverges.

(f) Let $a_n = \frac{1}{n^{1+\frac{1}{n}}}$, and $b_n = \frac{1}{n}$. Both a_n and b_n are positive for all $n \geq 1$.

$$
\begin{aligned}
\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{1}{n^{1+\frac{1}{n}}}\cdot\frac{n}{1}\\
&= \lim_{n\to\infty}\frac{1}{n^{\frac{1}{n}}}\\
&= 1,
\end{aligned}
$$

since $\lim_{n\to\infty} n^{1/n} = 1$ (see Tutorial 7 Question 4(x)).

The series $\sum_{n=1}^{\infty} b_n$ diverges since it is the Harmonic series. By the Limit Comparison Test, the series $\sum_{n=1}^{\infty} a_n$ also diverges.

(g)
$$\frac{e^n+1}{ne^n+1} \geq \frac{e^n+1}{ne^n+n} = \frac{e^n+1}{n(e^n+1)} = \frac{1}{n} \quad \text{for all} \quad n \geq 1.$$

By Comparison Test, the series $\sum_{n=1}^{\infty} \frac{e^n+1}{ne^n+1}$ diverges since the Harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

(h) Let $a_n = \dfrac{\sqrt{1+n}}{2+n}$, and $b_n = \frac{1}{\sqrt{n}}$. Both a_n and b_n are positive for all $n \geq 1$.

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{\sqrt{1+n}}{2+n} \cdot \frac{\sqrt{n}}{1} \\
&= \lim_{n\to\infty} \sqrt{\frac{n+n^2}{(2+n)^2}} \\
&= \lim_{n\to\infty} \sqrt{\frac{n+n^2}{4+4n+n^2}} \\
&= \lim_{n\to\infty} \sqrt{\frac{\frac{1}{n}+1}{\frac{4}{n^2}+\frac{4}{n}+1}} \\
&= \sqrt{\frac{0+1}{0+0+1}} = 1.
\end{aligned}$$

Since the series $\sum_{n=1}^{\infty} b_n$ diverges (it is a p-series with $p = 1/2 \leq 1$), by the Limit Comparison Test, the series $\sum_{n=1}^{\infty} a_n$ also diverges.

(i) Note that $4^n \geq n$ for all $n \geq 1$. So
$$0 \leq \frac{n+4^n}{n+6^n} \leq \frac{4^n+4^n}{n+6^n} \leq \frac{2\cdot 4^n}{6^n} = 2\left(\frac{4}{6}\right)^n.$$

The series $\displaystyle\sum_{n=1}^{\infty} \left(\frac{4}{6}\right)^n$ is a convergent geometric series. So the series $\displaystyle\sum_{n=1}^{\infty} 2\left(\frac{4}{6}\right)^n = 2\sum_{n=1}^{\infty} \left(\frac{4}{6}\right)^n$ also converges. By the Comparison Test, the series $\displaystyle\sum_{n=1}^{\infty} \frac{n+4^n}{n+6^n}$ also converges.

(j) Notice that $0 \leq \tan^{-1} n \leq \frac{\pi}{2}$ for all $n \geq 1$. Thus,

$$0 \leq \frac{\tan^{-1} n}{n^{1.2}} \leq \frac{\pi/2}{n^{1.2}}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^{1.2}}$ converges since it is a p-series with $p = 1.2 > 1$. It follows that the series $\sum_{n=1}^{\infty} \frac{\pi/2}{n^{1.2}} = \frac{\pi}{2} \sum_{n=1}^{\infty} \frac{1}{n^{1.2}}$ converges. By the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.2}}$ also converges.

$\square$

3. Find the values of p for which the series is convergent:

$$\sum_{n=1}^{\infty} n(1+n^2)^p.$$

Solution We will use the Limit Comparison Test. Notice the term $n(1+n^2)^p$ is very close to $n(n^2)^p = n^{2p+1} = \frac{1}{n^{-2p-1}}$. Set $a_n = n(1+n^2)^p$ and $b_n = \frac{1}{n^{-2p-1}}$. Then

$$\begin{aligned} \lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{n(1+n^2)^p}{n^{2p+1}} \\ &= \lim_{n\to\infty} \left(\frac{1+n^2}{n^2}\right)^p \\ &= \lim_{n\to\infty} \left(\frac{1}{n^2}+1\right)^p = 1. \end{aligned}$$

So by the Limit Comparison Test, either both $\sum b_n$ and $\sum a_n$ converges or both diverges. Since $\sum b_n$ converges if and only if $-2p-1 > 1$, we deduce thast $\sum a_n$ converges if and only if

$$\begin{aligned} &-2p-1 > 1 \\ \iff &-2p > 2 \\ \iff &p < -1. \end{aligned}$$

$\square$

Remark: Solving this question using the Integral Test will be more complicated.

4. (The Limit Comparison Test) Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms. Prove that if $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Solution Let M be a positive number. Since $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$, there exists an integer N such that

$$\frac{a_n}{b_n} > M, \quad \text{for all} \quad n > N,$$

or equivalently,

$$a_n > Mb_n \quad \text{for all} \quad n > N.$$

Since $\sum b_n$ diverges, the series $\sum Mb_n$ also diverges. By the Comparison Test, the "larger" series $\sum a_n$ also diverges. $\square$

5. Suppose $b_n > 0$ and $\sum_{n=1}^{\infty} b_n$ is convergent. Prove that the following series is also convergent.

(i) $\sum_{n=1}^{\infty} \ln(1 + b_n)$

(ii) $\sum_{n=1}^{\infty} \sin(b_n)$

Hint: Use the Limit Comparison Test, and the l'Hospital's Rule.

Solution

(i) Let $a_n = \ln(1 + b_n)$. Both a_n and b_n are positive for all $n \geq 1$.

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{\ln(1+b_n)}{b_n} \\
&= \lim_{x\to 0} \frac{\ln(1+x)}{x} \quad (\textstyle\sum b_n \text{ converges} \Longrightarrow b_n \to 0) \\
&= \lim_{x\to 0} \frac{1}{1+x} \quad \text{(L'Hospital Rule)} \\
&= 1.
\end{aligned}$$

By the Limit Comparison Test, the series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \ln(1 + b_n)$ converges.

(ii) Let $a_n = \sin(b_n)$. Since $\sum b_n$ is convergent, we have $\lim_{n\to\infty} b_n = 0$. Thus, there exists N such that $n > N \Longrightarrow \sin(b_n) > 0$. So for $n \geq N+1$, both a_n and b_n are positive.

We can use the Limit Comparison Test for $n \geq N+1$.

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{\sin(b_n)}{b_n} \\
&= \lim_{x\to 0} \frac{\sin x}{x} \\
&= \lim_{x\to 0} \frac{\cos x}{1} \quad \text{(L'Hospital's Rule)} \\
&= \cos 0 = 1.
\end{aligned}$$

By the Limit Comparison Test, the series $\sum_{n=N+1}^{\infty} a_n$ converges. It follows that the series $\sum_{n=1} a_n$ also converges.

Answer Keys. **1.** (a) converges (b) diverges (c) converges **2.** (a) converges (b) diverges (c) converges (d) converges (e) diverges (f) diverges (g) diverges (h) diverges (i) converges (j) converges **3.** $p < -1$